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GenEEG: Improving epileptic EEG detection through patient-adaptive latent diffusion and continual learning

Soinik Ghosh, Shiru Sharma, Neeraj Sharma  [Show more](#) 

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Highlights

- Proposes GenEEG, a VAE-LDM framework using 12 neurophysiological biomarkers to synthesize patient-specific EEG signals.
- Introduces CL-LOPO validation protocol with strict temporal separation to prevent data leakage across patients.
- Combines Elastic Weight Consolidation with Experience Replay to prevent catastrophic forgetting in sequential learning.
- Achieves Macro F1-score of 0.84 on adult and pediatric datasets, outperforming baselines by 15 percentage points.
- Provides open-source code and transparent analysis of high-frequency spectral limitations in the VAE architecture.

Abstract

Automated seizure detection systems face significant challenges due to the limited availability of clinical EEG data, a substantial class imbalance between seizure and non-seizure recordings, considerable variability among patients, and the issue of catastrophic forgetting in sequential

multi-patient learning. These issues greatly limit the effectiveness of machine learning models in monitoring and predicting epileptic seizures in clinical settings. We introduce GenEEG, a continual learning framework that combines neurophysiologically conditioned variational autoencoders (VAE) with latent diffusion models (LDM) for generating synthetic EEG data that adapts to patients in class-imbalanced seizure detection.

GenEEG includes three major contributions: (1) a dual-conditioned VAE-LDM that allows precise control over synthetic EEG using clinical states and 12 neurophysiological features, improving classification metrics; (2) a continual learning leave-one-patient-out (CL-LOPO) validation protocol with fold-specific normalization to prevent test-set leakage; and (3) a hybrid approach to stop catastrophic forgetting that combines Elastic Weight Consolidation with experience replay for adapting to sequential patients.

Rigorous evaluation on Siena Scalp EEG (adult) and CHB-MIT (pediatric) datasets shows significant performance gains with GenEEG, achieving macro F1-scores of 0.84 and 0.82 on both datasets, with GenEEG-augmented classifiers demonstrating consistent improvements over traditional oversampling baselines by 15 percentage points (F1: 0.84 vs. 0.69), while maintaining ictal sensitivity above 75 % across diverse populations. The continual learning approach offers memory efficiency benefits (4.8 GB vs. 12.4 GB for pooled training), though full training remains computationally intensive. While the framework excels at low-frequency capture (<30 Hz), high-frequency (>30 Hz) fidelity is limited by the $8\times$ compression architecture.

Ablation studies and statistical tests indicate that the majority of neurophysiological feature distributions in the generated data are similar to those observed in real recordings. GenEEG thus offers a reproducible, clinically relevant approach to address data scarcity and improve generalizable seizure detection systems.

Introduction

Epilepsy is a long-term neurological disorder characterized by repeated, unexpected seizures. These are sudden bursts of abnormal electrical activity in the brain that can cause convulsions, changes in consciousness, or subtle behavioral shifts [1]. The unpredictable and brief nature of seizures makes them hard to detect and diagnose. Symptoms can vary widely, from dramatic physical events to subtle sensory or cognitive changes [2]. Electroencephalography (EEG) is the main tool for detecting and studying epilepsy. It provides a non-invasive and detailed view of brain activity. However, collecting large, high-quality EEG datasets is a significant challenge. This is especially true for capturing rare preictal and ictal events due to the unpredictable nature of seizures and the variability among different patients. The lack of comprehensive data, along with imbalances in class distribution and the complicated nature of EEG signals, restricts the effectiveness of machine learning models for automatically detecting and predicting seizures [3], [4], [5].

In recent years, significant progress has been made in applying machine learning and deep learning techniques to EEG-based seizure detection. Traditional methods typically rely on handcrafted features, such as energy, entropy, or wavelet-based measures, which are then classified using standard machine learning algorithms [6], [7], [8]. However, these approaches depend on expert-driven feature extraction and may not perform well across different patient groups [9], [10].

Deep learning models, especially convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid frameworks, have shown better performance. They learn hierarchical representations directly from raw or lightly processed EEG data [11], [12], [13], [14]. Hybrid methods that combine graph neural networks (GNNs) with deep recurrent units have further enhanced seizure detection by capturing both spatial and temporal patterns in EEG signals, achieving high accuracy and reliability across various datasets [15], [16], [17]. Techniques for optimizing and improving model performance have also been suggested, addressing issues like high computational demands and slow convergence [18], [19].

Despite these advancements, the limited availability of labeled EEG data, particularly for rare seizure events, remains a major obstacle. Traditional data augmentation methods, such as signal transformations or noise addition, often miss the subtle, patient-specific features of epileptic EEG [20], [21]. To overcome this, deep generative models have gained traction as powerful tools for creating realistic EEG data. Generative Adversarial Networks (GANs) and their variations, such as WGAN-GP and DCGAN-GP, have successfully augmented EEG datasets and enhanced classifier performance, particularly in cases involving imbalanced or limited sample sizes [22], [23], [24]. Conditional GANs (CGANs) and models that focus on attention also boost the diversity and quality of synthetic data, leading to better generalization for subsequent tasks [25], [26], [27].

Variational Autoencoders (VAEs) provide steady training and effective learning of latent representations. This allows for the generation of synthetic EEG that closely resembles the statistical properties of real data [28], [29]. However, their success often requires careful tuning of loss functions and adequate data availability. More recently, Denoising Diffusion Probabilistic Models (DDPMs) and their latent variants have demonstrated promise in generating high-quality, diverse EEG signals, supporting data augmentation and addressing the issue of data scarcity [30], [31]. These models can capture complicated patterns over time and frequency, but they are resource-intensive and require careful management between data quality and generation speed [32].

Hybrid generative approaches are also gaining popularity. These combine different architectural strengths, such as integrating reinforcement learning with generative models or merging GNNs with GANs to further enhance the accuracy and usefulness of synthetic EEG data for clinical applications [33], [34]. Recent studies emphasize the importance of explainable AI (XAI), as interpretability and transparency are crucial for gaining clinician trust and ensuring the successful implementation of AI in real-world settings [35], [36], [37].

Despite these hopeful advances, several research gaps remain. Many generative models are tested on limited datasets or within specific frameworks, raising concerns about their effectiveness across varied patient groups, seizure types, and EEG systems [38]. The absence of standardized benchmarks and multi-center validation hinders the establishment of best practices [39]. Additionally, the ultimate goal for synthetic EEG data goes beyond classification; it aims to facilitate the creation of high-fidelity generative models, which can serve as statistically plausible facsimiles of patient data to aid in the development of personalized epilepsy treatment strategies [40], [41].

In this study, we present GenEEG, a generative model with a continual learning framework that addresses key challenges in EEG-based seizure detection through novel technical contributions that

extend beyond the application of existing methods (Fig. 1). GenEEG addresses data scarcity through neurophysiologically-conditioned synthetic EEG generation. The primary objective is to improve downstream classifier performance on imbalanced datasets for patient-specific seizure detection tasks. The main technical contributions are as follows:

1. Neurophysiological Feature-Conditioned Latent Diffusion for EEG: While latent diffusion models exist for natural images, we are the first to develop dual-signal conditioning for generating epileptic EEG. Our LDM works with both class labels (Normal/Preictal/Ictal) and 12-dimensional neurophysiological feature vectors. These vectors include spectral band powers (delta, theta, alpha, beta, gamma), complexity metrics (Higuchi fractal dimension and Hjorth parameters), entropy measures (spectral entropy and permutation entropy), and statistical moments (kurtosis). This allows us to control the clinical properties of generated signals beyond just their class membership, enabling us to target specific seizure morphologies.

2. Clinical-Ready Continual Learning Protocol (CL-LOPO): We create the first thorough validation framework that combines continual learning with leave-one-patient-out evaluation for biomedical AI research. Traditional LOPO methods often face data leakage during normalization. Our protocol prevents this by ensuring complete patient isolation through fold-specific preprocessing and temporal separation. This allows us to evaluate both cross-patient generalization and the effectiveness of sequential learning.

3. Dual Catastrophic Forgetting Prevention for Sequential Patient Learning: We merge Elastic Weight Consolidation (EWC) with Experience Replay (ER) in a unified framework tailored for patient-sequential epilepsy data. EWC safeguards critical network parameters in *both* the VAE and LDM using patient-specific Fisher Information Matrices computed after each patient's training, while ER keeps representational diversity intact by strategically buffering 500 samples per patient. We apply differentiated regularization strengths ($\lambda_{EWC}^{VAE} = 10000$, $\lambda_{EWC}^{LDM} = 5000$) to account for model complexity differences and ensure stable continual learning. This dual mechanism protects against forgetting while allowing efficient memory use for clinical applications.

The framework includes a two-stage VAE-LDM architecture optimized for multi-channel EEG dynamics, achieving an $8\times$ latent compression factor. It utilises a universal 16-channel selection based on the International 10–20 system, ensuring compatibility across datasets without requiring changes to the architecture. This design offers memory efficiency benefits with 4.8 GB peak usage compared to 12.4 GB for pooled training ($2.6\times$ reduction), though full training remains computationally intensive. Together, these innovations enable GenEEG to generate synthetic EEG data that closely resembles real recordings both statistically and physiologically. This significantly enhances the reliability of seizure detection models.

The remainder of this paper is structured as follows. Section 2 details the proposed methodology, covering how we prepared the datasets, designed the generative model, and established the CL-LOPO evaluation framework. Section 3 presents the experimental results, analyses, and statistical validations. Section 4 discusses interpretability, computational implications, and clinical implications. Finally, Section 5 concludes the paper with key insights and recommendations for future research in large-scale, real-time clinical use.

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Section snippets

Methods

To tackle the issue of limited data and class imbalance in seizure analysis, we created a deep generative framework to synthesize high-quality, multi-channel electroencephalogram (EEG) signals. Our approach features a two-stage generative model that is trained and tested using a patient-adaptive continual fine-tuning method. This helps us evaluate both the quality of the generation and the model's ability to work with new, unseen patient data. All experiments were conducted in PyTorch [42] with ...

Results

We evaluated GenEEG using strict CL-LOPO cross-validation on the Siena dataset, comprising 14 patients, and the CHB-MIT dataset, comprising 24 patients. We took measures to prevent temporal leakage. The results are organized as follows: (1) Cross-dataset validation comparing continual learning with pooled training baselines, showing the framework's efficiency and scalability; (2) A quantitative assessment of the quality of synthetic EEG through statistical feature analysis and latent space ...

Discussion

In this work, we introduce GenEEG, a generative framework designed to adapt to patients. It tackles the challenges of limited data and class imbalance in analyzing epileptic EEGs. We achieve this through continuous learning and by preventing temporal leaks. Our thorough evaluation reveals that this two-stage VAE-LDM architecture generates realistic synthetic EEG and enhances diagnostics under stringent cross-patient validation protocols.

Key Technical Contributions: The success of our framework ...

Conclusion and future work

This work presents **GenEEG**, a patient-adaptive generative framework that combines neurophysiologically informed latent representation learning with conditional diffusion modeling under a continual-learning evaluation protocol. By linking a multi-objective VAE with a feature-conditioned latent diffusion model and embedding these within a CL-LOPO scheme, which promotes seizure-aware temporal separation and fold-specific normalization, GenEEG offers a detailed and clinically relevant assessment of ...

CRediT authorship contribution statement

Soinik Ghosh: Writing – original draft, Visualization, Validation, Methodology, Conceptualization.

Shiru Sharma: Writing – review & editing, Supervision, Conceptualization. **Neeraj Sharma:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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Declaration of generative AI/Generative AI

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